FORM TP 2023017

TEST CODE **01229020**

JANUARY 2023

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

INFORMATION TECHNOLOGY

Paper 02 – General Proficiency

2 hours

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of FOUR questions. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
5. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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"*"Barcode Area"*"
Sequential Bar Code

Answer ALL questions.

1. (a) Complete the table provided below by identifying EACH of the following application software types which matches the descriptions shown in the table.

General-purpose Special-purpose

Custom-written Customized

Application Software Type	Description
	Software developed to perform a specific task rather than a range of functions, for example, online gaming software for a specific game.
	Software developed for use in specific organizations, for example, for the military, for banks, for air traffic control or for medical equipment.
	Software not specific to any organization or company and the user decides what to use the software for.
	Software that has been modified by the developer to perform a specific task for the user.

(3 marks)

(b) (i) Explain what is meant by the following statement.

'A turnaround document is a human-readable document that is transformed into a machine-readable document.'

.....

.....

.....

(2 marks)

(ii) Outline ONE example of a turnaround document used in an application as described in the statement in (b) (i).

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(2 marks)

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(c) The following table shows the actions taken by a user entering a URL in the address bar of a web browser in order to display a web page on the computer.

Number the actions in the table to show the correct order of the steps needed to display the web page.

Actions	Step No.
The web browser obtains the IP address of the hosting web server.	
The web browser interprets and displays the HTML content.	
The web browser sends a HTTP requests to the web server.	
The web browser parses the URL to obtain the domain name.	

(3 marks)

(d) The following extract is a Table of Contents showing the pages for different sections of a document.

Contents

Birdsong Pan Grove 2

Pan Practice 2

Practice Schedules 2

Red Devils Football Academy 3

Registration Form 3

Disclaimer 3

(i) Identify the TWO heading styles used in the Table of Contents.

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.....

(2 marks)

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(ii) State, in the correct order, the steps required to apply the heading styles in the Table of Contents.

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(2 marks)

(iii) State, in the correct order, the FOUR steps required to create the Table of Contents on the **first** page of the document.

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(4 marks)

(iv) Explain how you would go from the Table of Contents directly to the page with the heading entitled "Registration Form".

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(2 marks)

Total 20 marks

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2. (a) Complete the following table with FOUR **contrasting** differences between an Intranet and the Internet.

	Intranet	**Internet**
1.		
2.		
3.		
4.		

(4 marks)

(b) On a data capture form, a user enters passport number and age. Identify ONE appropriate type of validation check which can be used for EACH entry, giving a reason.

Passport number:

.....

Age:

.....

.....

(4 marks)

(c) Complete the following table with the missing description and term relating to computer misuse.

Description	**Term**
Acquiring sensitive information, for malicious reasons by deceiving the user through the content of an email	
	Hacking

(2 marks)

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(d) Figure 1, Quarter1Sales, is a worksheet which shows a summary of the total amount sold by each salesperson and Figure 2 is a 1-D PivotTable created from Figure 1.

Figure 3 shows the fields from the worksheet Quarter1Sales in Figure 1 that were dragged into the different areas of the PivotTable Fields dialogue box to produce the 1-D PivotTable in Figure 2.

	A	B	C	D
1	Salesperson	Region	Order Amount	Month
2	Albert, Carol	East	$925.00	January
3	Albert, Carol	East	$875.00	February
4	Albert, Carol	East	$350.00	March
5	Als, Michael	West	$1,500.00	January
6	Als, Michael	West	$550.00	February
7	Als, Michael	West	$400.00	March
8	Rampersad, John	South	$235.00	February
9	Rampersad, John	South	$850.00	January
10	Rampersad, John	South	$600.00	March
11	Zepyrin, Richard	West	$400.00	January
12	Zepyrin, Richard	West	$965.00	February
13	Zepyrin, Richard	West	$125.00	March
14	Hernades, Erica	South	$1,500.00	March
15	Hernades, Erica	South	$1,350.00	January
16	Hernades, Erica	South	$435.00	February
17				

Sheet1 Quarter1Sales

Figure 1. Quarter1Sales

	A	B	C	D
1	Row Labels	Sum of Order Amount		
2	Albert, Carol	2150		
3	Als, Michael	2450		
4	Hernades, Erica	3285		
5	Rampersad, John	1685		
6	Zepyrin, Richard	1490		
7	Grand Total	11060		

Figure 2. 1-D PivotTable

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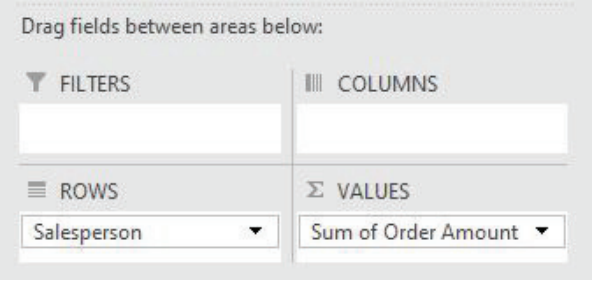

Figure 3. PivotTable Fields Dialogue Box

(i) Describe the change you would make to the PivotTable Fields Dialogue Box in Figure 3, to show a summary of the total amount sold by EACH region.

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(2 marks)

(ii) Describe the change you would make to the PivotTable Fields Dialogue Box in Figure 3 to create the following.

A 2-D PivotTable showing a summary with a column for EACH salesperson's monthly sales (January, February and March) **and** the salesperson's total amount sold.

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(2 marks)

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(iii) Write the formula for Cell C17 of the worksheet Quarter1Sales to calculate the total of **all** order amounts in Column C.

.....

.....

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.....

(2 marks)

(iv) Link the 1-D PivotTable in Figure 2 with the worksheet Quarter1Sales in Figure 1, by writing a formula for Cell C1 of the PivotTable, to show the sum of all order amounts created for Cell C17 in (iii).

.....

.....

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.....

(2 marks)

(v) Write a formula for Cell C2 in Figure 2, to be replicated through to Cell C7, to calculate what percentage of the total order amounts in Cell C1 is the salesperson's order amount in B1. (Percentage is calculated by salesperson's order amount divided by total order amounts multiplied by 100.)

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(2 marks)

Total 20 marks

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3. (a) A relational database is composed of the following four objects, that all work together with the same data.

Tables Forms Queries Reports

Complete EACH of the statements below using the **most** appropriate term from the list above. (Any term may be used more than once.)

(i) A relational database stores data in a number of related files, more commonly referred to as ..... .

(ii) ..... are used to ask detailed questions of the database when searching for records that satisfy particular conditions.

(iii) ..... display fields extracted from ..... in a particular order.

(iv) ..... use a graphical representation to allow users to enter, modify, delete and view records in the database. **(5 marks)**

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Figures 4 and 5 are two database tables named CUSTOMERS and ORDERS respectively.

Figure 6 is the dialogue box showing the relationship between the CUSTOMERS and ORDERS tables.

Customers

ID	First Name	Last Name	Street Address	Phone Number	City
1	Whitney	Thurab	7 Walker Drive	868-330-2314	Diego Martin
2	Lucy	Patrick	7 Brewster Street	868-555-4534	Morvant
3	Jerad	Jarvis	2 St. Augustine Avenue	868-333-4564	Diego Martin
4	Brendan	King	47 Hillbury Street	868-333-7653	Diego Martin
5	Charlene	Jones	23 Solomon Lane	868-333-8658	Diego Martin
6	Quinton	De Kock	4 Cypress Cresent	868-234-5112	Carenage
7	Alexander	Hinkson	10 Hodgkinson Drive	868-555-5460	Carenage
8	Nisha	Cumberbatch	123 Hunter Street	868-334-5753	Diego Martin
9	Hillary	Morris	25 Newman Drive	868-335-9745	Diego Martin
10	Kyara	Griffith	14 Miller Jones Road	868-555-8975	Carenage

Figure 4. Customers Table

Orders | Order Items | Customers

ID	Customer ID	Paid	Pre Order	Pickup Date	Click to Add
5	1	Yes	No	26/03/2014	
6	2	Yes	Yes	04/02/2014	
7	13	Yes	No	20/02/2014	
8	15	Yes	Yes	19/02/2014	
9	14	Yes	No	18/03/2014	
10	12	Yes	No	17/06/2014	

Figure 5. Orders Table

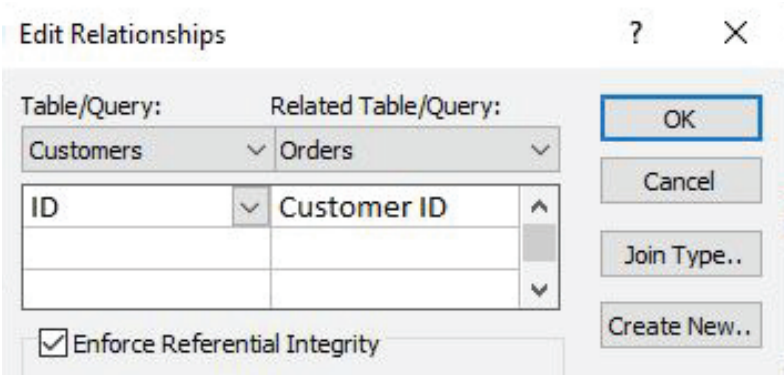

Figure 6. Dialogue Box showing the relationship between Customers and Orders tables.

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(b) The entries in the dialogue box in Figure 6 **on page 12** are used when joining the database tables, CUSTOMERS and ORDERS, to establish a **one to many** relationship.

Describe EACH of the following entries.

(i) ID

.....

(1 mark)

(ii) Customers ID

.....

(1 mark)

(iii) Enforce Referential Integrity

.....

.....

.....

(2 marks)

(c) Using the following Query Design window, create a query to extract the fields ID, First Name, Last Name, Phone Number and City from the CUSTOMERS table for all customers from the city of Carenage and whose phone numbers are greater than 868-500-000.

Field	ID	First Name	Last Name	Phone Number	Street Address	City
Table	Customers	Customers	Customers	Customers	Customers	Customers
Sort						
Show	☐	☐	☐	☐	☐	☐
Criteria						
OR						

(4 marks)

(d) State the ID(s) of the record(s) extracted when the query designed in (c) is executed.

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(2 marks)

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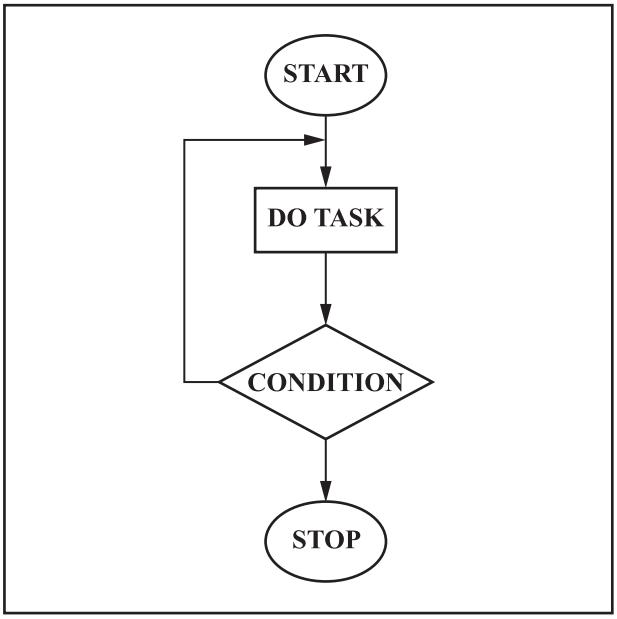
(e) (i) The following flowchart shows a REPEAT UNTIL loop. Complete the flowchart by labelling the TWO directional lines leaving the condition box as TRUE and FALSE.

REPEAT UNTIL LOOP



(2 marks)

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(ii) Draw a flowchart of the WHILE DO ... ENDWHILE loop in the box provided below. **(3 marks)**

WHILE DO ... END WHILE LOOP

(f) (i) Complete the table below based on the statements in it and the flowcharts in (e) (i) and (ii).

Statement	WHILE DO...END WHILE Loop	REPEAT UNTIL Loop
Is the condition tested at the START or END of the loop?		
Does the loop executes as long as the condition is TRUE or FALSE?		
Statements in the loop must be executed: YES or NO?		
Number of times to repeat the loop is known: YES or NO?		

(4 marks)

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(ii) State the MAIN requirement for the use of the FOR loop.

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(1 mark)

Total 25 marks

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4. (a) An algorithm is needed to read the age of each student until it reads an age of 999. It should then calculate the average age and output both the number of students and the average age of all students.

Complete the IPO Chart below to define the problem by decomposing it into EACH of its significant components: Input, Processing and Output.

Input	**Processing**	**Output**

(5 marks)

(b) The following is the pseudocode for the algorithm mentioned in (a).

```
START
Set COUNT ← 0
Set AGE ← 0
Set AVERAGE ← 0
Set TOTAL ← 0
Write "Enter Student's age"
Read AGE

WHILE AGE <> 999 DO
        Set TOTAL ← TOTAL + COUNT
        Set COUNT ← COUNT + 1
        Write "Enter another age or 999 to terminate the program"
        Read AGE
ENDWHILE
        Set AVERAGE ← TOTAL / COUNT
        Write "Number of Students:" , COUNT
        Write "Average age of Students: " , AVERAGE
STOP
```


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(i) Test the pseudocode for correctness by completing the entries in the trace table below for EACH loop of the algorithm when the algorithm reads EACH of the following AGE values: 18, 26, 39 and 999.

AGE	TOTAL	COUNT	AVERAGE
18	0		0
26		2	
39			0
999	2	3	

(6 marks)

(ii) Write the TWO output lines produced by the algorithm after the dry run in (b) (i).

.....

.....

(2 marks)

(iii) Identify the line in the pseudocode in (b) **on page 17** that needs to be corrected and state the correction to the line.

Line: .....

Correction: .....

(2 marks)

(c) Identify TWO impacts of Information Communication Technology (ICT) on the work pattern of employees in the health sector.

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.....

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(2 marks)

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(d) The workplace has changed drastically because of the ICT revolution. State ONE positive and ONE negative aspect of automation on skilled and unskilled jobs.

Positive: .....

Negative: .....

(2 marks)

(e) For EACH of the following files, state the MOST appropriate file organization method and give a reason for EACH choice.

File A: Meter reading records submitted by customers monthly
File B: Personal data of customers in a master file
File C: Customers' usernames and passwords

File	Organization Method	Reason
A	.....	....
B	.....	....
C	.....	....

(6 marks)

Total 25 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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EXTRA SPACE

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